FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: COMMERCIAL ARITHMETIC 1	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

You have now completed all eight lessons in the Commercial Arithmetic 1 unit. In this Final Explanation, you will use everything you have learned — from budgeting and discounts to commissions, profit and loss, and currency conversion — to answer the driving question: How can Amina — and every Kenyan — use commercial arithmetic to make smart, fair, and informed money decisions in markets, clubs, and across borders? Write your answer in the five sections below. For each section, refer back to your Summary Table, your lesson notes, and the models you built and revised throughout the unit. Your answer must be written in your own words and must connect clearly to Amina's story and to real Kenyan contexts (markets, mobile money, cross-border trade, school clubs, and everyday buying and selling). Use correct mathematical vocabulary (e.g., marked price, discount, commission rate, cost price, selling price, profit, loss, exchange rate, buying rate, selling rate). Where you make a mathematical claim, support it with a worked calculation or a specific numerical example. Your complete response across all five sections must be a minimum of 300 words. Strong responses will be longer, more detailed, and will show deep understanding of how the different topics connect to each other and to Amina's real financial decisions.

SECTION 1: BUDGETING — PLANNING AMINA'S KSh 4,500

Explain what a budget is and why it is necessary. Using Amina's KSh 4,500 as your example, show how she should divide her money across the three goals (Environmental Club trip contribution, smartphone purchase, and the Uganda transfer) before she spends a single shilling. Explain what could go wrong if Amina did NOT make a budget first. Connect your answer to the key inquiry question: Why do we need a budget?	A budget is a written plan that shows how you will use your money before you spend it. It lists all the money you have coming in (income) and all the things you plan to spend it on (expenditure), so that your total spending does not exceed your total income. Without a budget, you might spend too much on one thing and have nothing left for something else that matters just as much. Amina starts with exactly KSh 4,500. Before she does anything else, she should write down her three goals and estimate what each one will cost. For example, she might decide to set aside KSh 800 for the Environmental Club trip contribution, KSh 3,200 for the smartphone purchase, and keep the remaining KSh 500 for the Uganda transfer — a total of KSh 4,500. This simple plan immediately shows her whether her goals are realistic or whether she needs to adjust them. If Amina skips the budget and goes straight to the Kondele electronics stall, she might spend more than she planned on the phone and then discover she has too little left to send anything meaningful to her aunt in Uganda. Budgeting is therefore not just a school exercise — it is the very first tool of smart money management, used by individuals, families, school clubs, and even the Kenyan national government every year when Parliament passes the Finance Bill.

SECTION 2: DISCOUNTS — THE REAL PRICE AT THE KONDELE STALL	
Explain what a discount and a percentage discount are. Use the smartphone scenario from Lesson 2 and Lesson 3 to show Amina (and any buyer) how to calculate the actual selling price after a discount, and how to compare two different discount deals to find the better value. Explain why simply hearing '20% off' is not enough information for a smart buyer. Connect your answer to the key inquiry question: Why are discounts and commissions necessary in trade?	A discount is the amount by which a seller reduces the original (marked) price of a product. A percentage discount tells you what fraction of the marked price is being taken off. To find the actual discount amount, you multiply the marked price by the percentage discount and divide by 100. The selling price is then the marked price minus the discount amount. Suppose the smartphone at the Kondele stall has a marked price of KSh 4,000 and is advertised at '20% off.' The discount amount is (20 ÷ 100) × 4,000 = KSh 800, so the selling price is KSh 4,000 − KSh 800 = KSh 3,200. This means Amina needs KSh 3,200, not KSh 4,000 — a real saving. However, if she only has KSh 3,000 set aside for the phone, the 20% discount still leaves her KSh 200 short, which is exactly the puzzle she faces in our phenomenon. Hearing '20% off' is not enough on its own because the saving depends entirely on the marked price. A 20% discount on a KSh 4,000 phone saves KSh 800, but a 25% discount on a KSh 3,600 phone saves only KSh 900 — so you need to compare actual selling prices, not just percentages. Discounts are necessary in trade because they attract customers, help sellers move stock quickly (for example, a Gikomba market trader selling end-of-season clothes), and reward buyers who pay promptly or buy in bulk. A smart buyer like Amina always calculates the final price before deciding which deal is better.

SECTION 3: COMMISSION — WHO GETS PAID WHEN AMINA SENDS MONEY?	
Explain what commission and percentage commission are, and why agents charge commission. Use the mobile-money transfer scenario (Amina sending money to her aunt in Uganda) to show how commission is calculated and how it affects the amount Amina needs to have or the amount her aunt actually receives. Explain the difference between a buyer paying commission and a seller earning commission. Connect your answer to the key inquiry question: Why are discounts and commissions necessary in trade?	Commission is a fee charged by an agent or middleman for providing a service — in this case, the mobile-money agent who processes Amina's transfer to Uganda. It is usually expressed as a percentage of the transaction amount. To calculate the commission, you multiply the transaction amount by the commission rate and divide by 100. For example, if Amina wants to send KSh 500 to her aunt and the mobile-money agent charges a 5% commission, the commission is (5 ÷ 100) × 500 = KSh 25. This means Amina must hand over KSh 525 in total, or alternatively her aunt receives only KSh 475 depending on who bears the fee — the sender or the receiver. Either way, the commission reduces the value of the transaction for the intended recipient, which is why Amina needs to factor it into her budget before she decides how much to set aside for the transfer. Commission is necessary in trade because agents provide a real and valuable service — they operate the transfer systems, take on risk, and make trade possible across distances and borders (for example, between Kisumu and Kampala). Without commission, agents would have no income and would not offer the service at all. The same principle applies to a commission-earning sales representative at a Nairobi electronics shop: the commission motivates the agent to work hard and complete transactions, which benefits both buyers and sellers. Understanding commission protects people like Amina from being surprised by hidden charges.

SECTION 4: PROFIT, LOSS, AND AMINA'S MANDAZI BUSINESS	
Explain the concepts of cost	The cost price (CP) is the total amount of money a trader spends to obtain or produce goods — for Amina, this includes flour, fat,

price, selling price, profit, and loss. Show how to calculate profit or loss as an amount and as a percentage. Use Amina's mandazi-selling business as your main example and include at least one worked calculation. Explain why understanding profit and loss matters for any young Kenyan who wants to run even a small business — at school or at home.	sugar, firewood, and any other costs for making a batch of mandazi. The selling price (SP) is the amount of money she receives when she sells those mandazi. If SP is greater than CP, Amina makes a profit; if CP is greater than SP, she makes a loss. Suppose Amina spends KSh 300 making a batch of mandazi and sells the whole batch for KSh 420. Her profit is KSh 420 − KSh 300 = KSh 120. To find the percentage profit, divide the profit by the cost price and multiply by 100: (120 ÷ 300) × 100 = 40%. Amina makes a 40% profit on that batch. Now suppose one week she sells the same batch for only KSh 270 because of low demand. Her loss is KSh 300 − KSh 270 = KSh 30, and the percentage loss is (30 ÷ 300) × 100 = 10%. Understanding profit and loss is essential for any young Kenyan entrepreneur. It tells you whether your business is actually working or whether you are losing money without realising it. Many small traders at markets like Wakulima in Nairobi or Kibuye in Kisumu price their goods by instinct and later discover their 'profit' was actually a loss once all costs are counted. By calculating profit and loss correctly, Amina can set the right price for her mandazi, know exactly how many batches she needs to sell to save KSh 4,500, and make better decisions about whether to expand or change her business.

SECTION 5: CURRENCY CONVERSION AND AMINA'S COMPLETE FINANCIAL PLAN	
Explain how currency conversion works, including the difference between the buying rate and the selling rate. Use the Kenya shilling (KES) to Uganda shilling (UGX) conversion in Amina's story as your main example and show at least one calculation. Then bring your whole answer together: describe Amina's complete financial plan, showing how budgeting, discounts, commission, profit/loss, and currency conversion all connect. Explain how commercial arithmetic empowers every Kenyan — not just Amina — to make smart, fair, and informed money decisions.	Currency conversion is the process of changing money from one country's currency into another. Every currency has an exchange rate — the price at which one currency can be bought or sold in terms of another. There are two rates to know: the buying rate (the rate at which a bank or agent buys foreign currency from you) and the selling rate (the rate at which the agent sells foreign currency to you). The selling rate is always slightly higher than the buying rate, and the difference is how the agent makes a profit on the transaction. For example, suppose the exchange rate on Monday is 1 KES = 28 UGX (Uganda shillings). If Amina sends KSh 500 to her aunt in Uganda, her aunt should receive 500 × 28 = 14,000 UGX — but remember, the mobile-money agent also charges a 5% commission (KSh 25), so Amina needs KSh 525 in total for her aunt to receive the full equivalent. If the exchange rate changes to 1 KES = 26 UGX by Tuesday, the same KSh 500 now converts to only 13,000 UGX. This shows why exchange rates must always be checked on the day of the transaction. Now you can see how Amina's complete financial plan brings all five topics together. She starts by making a budget — the foundation of every smart financial decision. She uses discount arithmetic to find the true price of the smartphone at the Kondele stall and to compare deals. She uses commission calculations to know the real cost of the mobile-money transfer. She uses profit and loss to understand how her mandazi business generated her KSh 4,500 in the first place, and to plan future savings. Finally, she uses currency conversion to ensure her aunt in Uganda receives the right amount. Each mathematical tool on its own is useful, but together they form a complete picture of financial literacy. Commercial arithmetic is not just a topic in a textbook — it is the mathematics of real life, empowering every Kenyan, from a student in Kisumu to a trader at Mombasa port, to make decisions that are smart, fair, and fully informed.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of	Excellent (4): Accurately and clearly	Proficient (3): Accurately defines and	Developing (1–2): Defines or explains

Commercial Arithmetic Topics	defines and explains all five topics (budgeting, discount, commission, profit/loss, and currency conversion) using correct mathematical vocabulary throughout. Demonstrates deep understanding of each concept with no errors.	explains at least four of the five topics using mostly correct vocabulary. Minor gaps or imprecise language do not affect overall understanding.	fewer than four topics, or contains significant conceptual errors (e.g., confuses selling price with marked price, or confuses profit with revenue). Mathematical vocabulary is largely missing or misused.
Mathematical Accuracy and Use of Calculations	Excellent (4): Every calculation presented is correct and clearly shown step by step (formula → substitution → answer with units). At least one fully worked calculation appears in each of Sections 2–5. All answers are in correct and appropriate units (KSh, %, UGX, etc.).	Proficient (3): Most calculations are correct and shown with working. At least three sections include a numerical example. One or two minor arithmetic slips are present but do not undermine the overall argument.	Developing (1–2): Calculations are frequently missing, incorrect, or presented without working. Fewer than two sections include a numerical example, making it difficult to assess mathematical understanding.
Connection to the Phenomenon (Amina's Story) and Kenyan Context	Excellent (4): Every section explicitly connects back to Amina's KSh 4,500 scenario and/or other specific Kenyan contexts (e.g., Kondele stall, Wakulima Market, Kibuye Market, Lake Victoria trip, mobile-money agent, Uganda shilling). Kenya-relevant examples feel natural and are integrated into the mathematical explanation, not just added as decoration.	Proficient (3): Most sections reference Amina or a Kenyan context. A few sections may rely on generic examples, but the overall response is clearly grounded in the phenomenon.	Developing (1–2): The response rarely or never references Amina or specific Kenyan contexts. The phenomenon connection is superficial or absent, and the answer reads as a general mathematics exercise.
Synthesis — Connecting All Topics into a Coherent Financial Plan	Excellent (4): Section 5 (and the response as a whole) clearly shows how budgeting, discounts, commission, profit/loss, and currency conversion work together as an integrated system. The student articulates why each tool is necessary and how failure to use any one of them could lead to a poor financial decision for Amina or any Kenyan.	Proficient (3): Section 5 attempts to bring the topics together and identifies at least three links between them. The synthesis is logical but may not fully explain how the tools interact as a system.	Developing (1–2): The response treats each topic in isolation with little or no attempt to connect them. Section 5 reads as a repetition of earlier sections rather than a synthesis.
Communication — Clarity, Structure, and Word Count	Excellent (4): All five sections are fully answered in clear, student-accessible language. Mathematical reasoning is easy to follow. The response meets or exceeds the 300-word minimum (across all sections combined) and is well organised with no significant grammatical or spelling errors that impede meaning.	Proficient (3): All five sections are attempted and the response meets the 300-word minimum. The writing is generally clear, though some explanations may be brief or slightly unclear. A few grammatical errors are present but do not impede understanding.	Developing (1–2): One or more sections are missing or left nearly blank. The response falls below the 300-word minimum, or the writing is so unclear or disorganised that the mathematical reasoning cannot be followed. Frequent errors impede meaning.